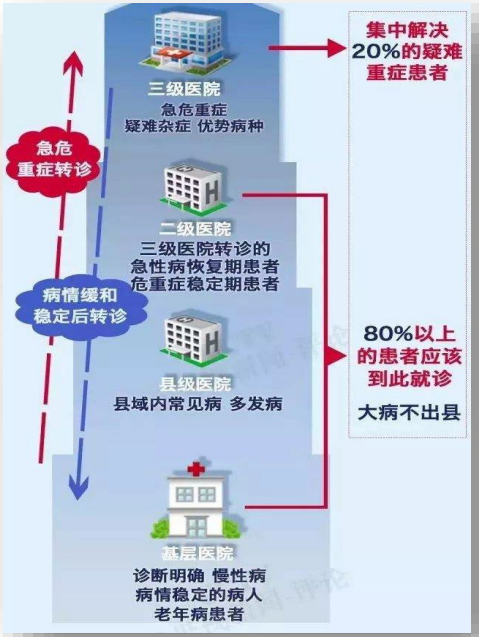
Stochastic Service Network Optimization for Resource Allocation in Medical Alliance

Supply and Demand Mismatch

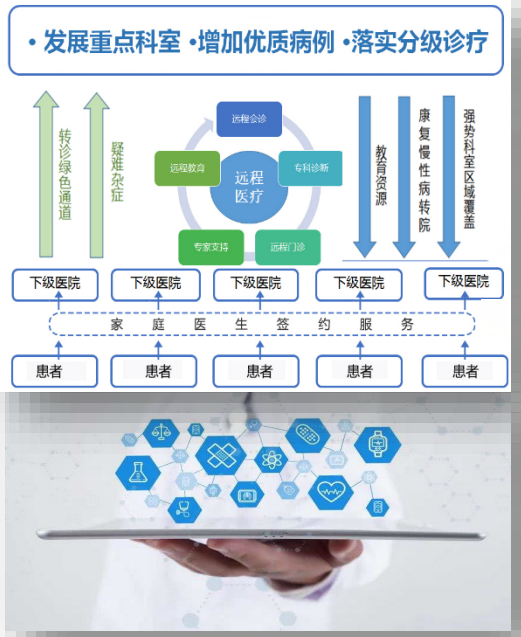
- Idle resources in primary medical institutes.
- Busy resources in general hospitals.
- Poor medical experience
- Low resource utilization.

Hierarchical Healthcare System



- Rational allocation of medical resources.
- Promotion of basic medical and health services equalization.

Medical Alliance System



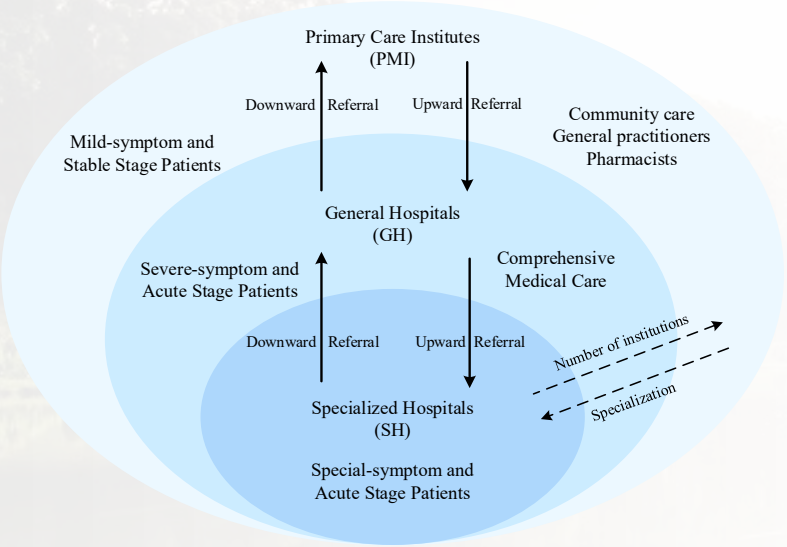
- Promotion of medical resource sinking.
- Promotion of patients two-way referral.

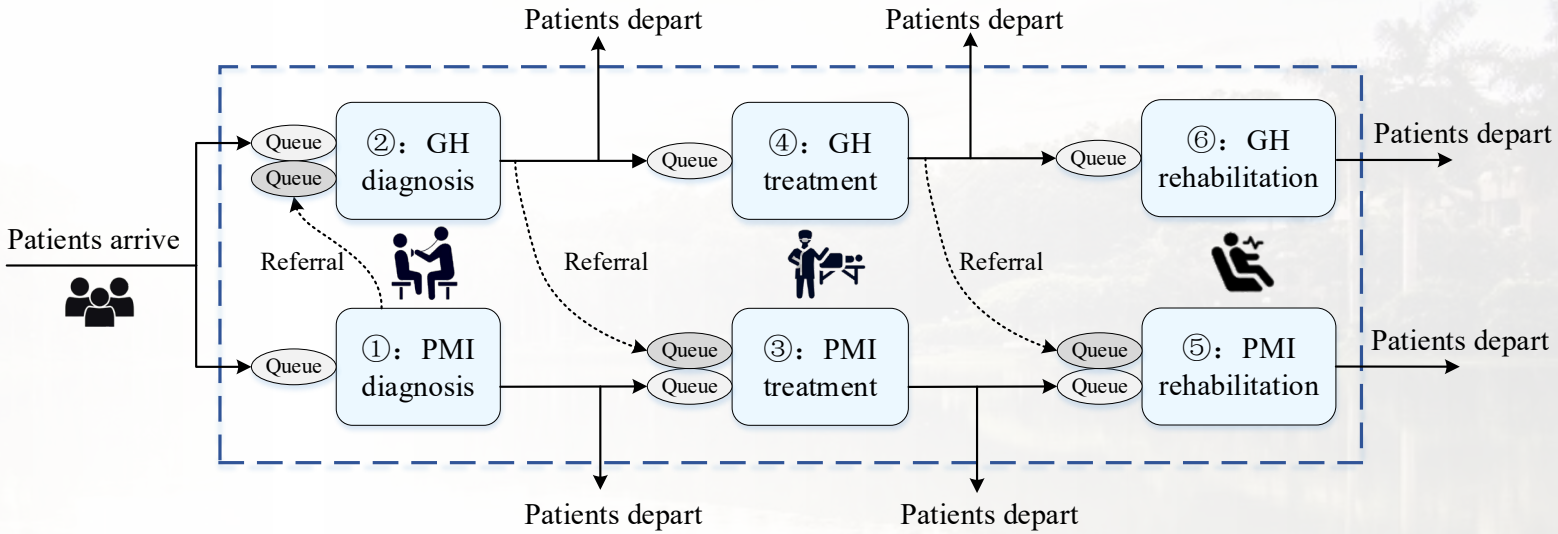
Resource Allocation

- Unreasonable allocation of medical resources among medical institutions

The Proposed Model

Medical Alliance Stochastic Service Network under Hierarchical Healthcare System





- **Theoretical Framework**: Stochastic Service Network
- **Medical Alliance**: Primary medical institutes; General hospitals
- **Patient Types**: Mild-symptom; Severe-symptom
- **Resource Types**: General medical resources; Quality medical resources
- **Medical Processes**: Diagnosis; Treatment; Rehabilitation
- **Stakeholders**: Medical alliance; Patients
- **Optimization Objectives**: System cost minimization; Patient utility maximization

The Proposed Model

Multi-objective Stochastic Service Network Optimization for Resource Allocation in Medical Alliance

➢ **Objective functions:**

- System cost minimization

Minimize: resource usage cost + resource idle cost + patient waiting cost + referral cost)

- Patient utility maximization

Maximize: patient perceived value + patient waiting cost + treatment cost + referral cost)

➢ **Constraints :**

- Service node in stable state
- Total Resource Constraints
- Resource adjustment rules:

  Quality medical resources will flow downward

  General medical resources will not flow upward

The Proposed Model

■ **Multi-objective Stochastic Service Network Optimization for Resource Allocation in Medical Alliance**

- System cost minimization

$$\min \sum_{i=1}^{6}\sum_{s=1}^{2}\left(x_{i,s}\rho_i c_{i,s,1} + x_{i,s}\left(1-\rho_i\right)c_{i,s,0}\right) + \sum_{i=1}^{6}\sum_{t=1}^{2} c_{i,t} l_{i,t} + c_f \sum_{i=1}^{6}\eta_{i,t}$$

- Patient utility maximization

$$\max \sum_{i=1}^{6} v_i \left(\sum_{t=1}^{2}\lambda_{i,t}\cdot\sum_{s=1}^{2} x_{i,s} c_{i,s,1}\right) - \sum_{i=1}^{6}\sum_{t=1}^{2} c_{i,t} l_{i,t} - c_f \sum_{i=1}^{6}\eta_{i,t} - \sum_{i=1}^{6}\sum_{t=1}^{2}\sum_{q=1}^{2} p_{i,q,t}\lambda_{i,q,t}$$

subject to

- Service node in stable state

$$0 \le \rho_i < 1$$

- Total Resource Constraints

$$\sum_{i=1}^{4} x_{i,1} = M_1$$

$$\sum_{i=5}^{6} x_{i,1} = N_1$$

$$\sum_{i=1}^{4} x_{i,2} \le M_2$$

$$\sum_{i=1}^{4} x_{i,2} \le N_2$$

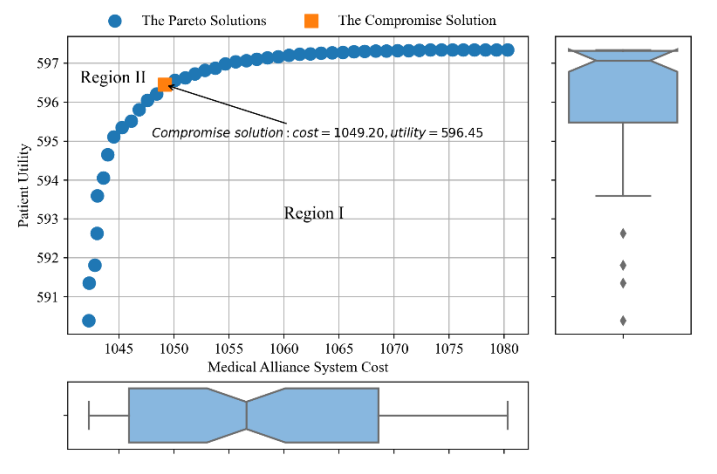
- Resource adjustment rules

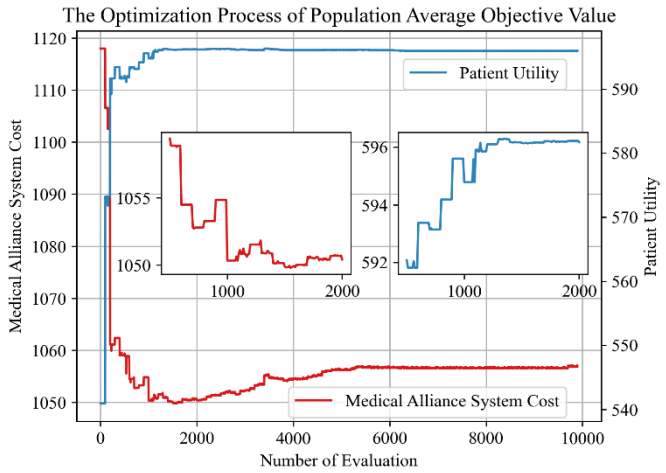
$$x_{i,2} \le x_{i,2},\ i = 2,4,6$$

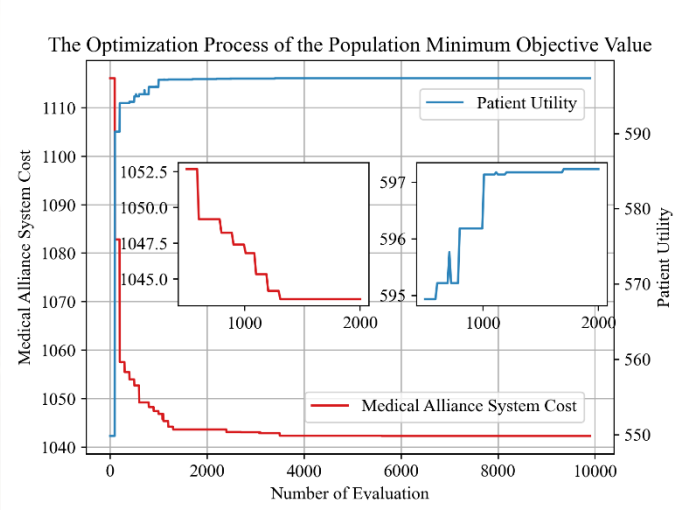
$$x_{i,s} \ge 0,\ x_{i,s} \text{ is an integer}$$

Simulation analysis

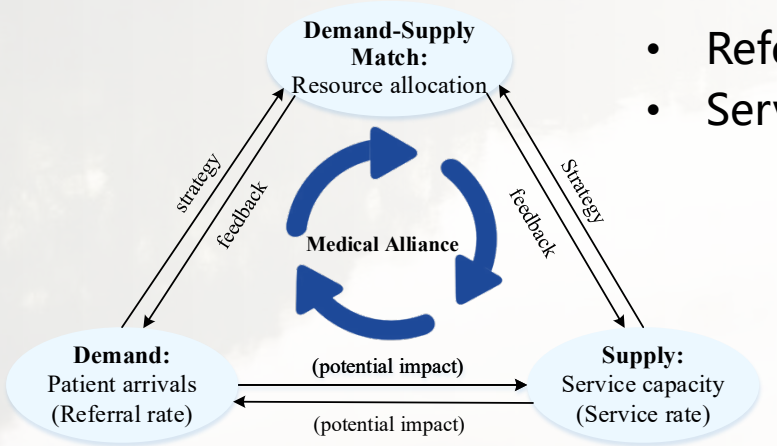
■ Multi-objective optimization (NSGAII) -> to obtain the optimal resource allocation strategies



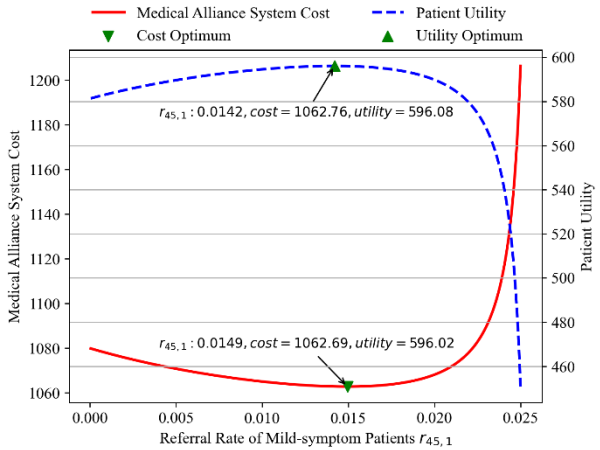


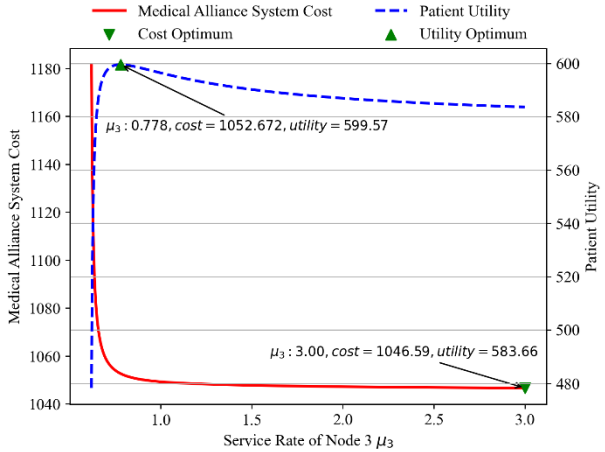


■ Optimization analysis: Influence of key strategies changes on resource allocation strategies



- Referral rate adjustment
- Service rate adjustment





Conclusions & Contributions

Conclusions

- Pareto front suggests that **the patient utility decreases as the medical alliance cost increases.** The choice of solution should be made with less system cost or more utility, depending on the specific decision context.
- **Centralized resource allocation** is most system cost effective in resource shortage situations and increases patient utility.
- **Both the referral rate and the rate of medical services affect the optimization goal of the medical association**, and decision makers should adjust them dynamically according to the actual situation.

Contributions

- A formal framework based on a **stochastic service network for allocating medical resources** is constructed, which shows how the medical alliance works under a referral system.
- A stochastic resource allocation problem is mathematically modeled to **reduce the overall system cost and improve the patients' utility**, and NSGAII is utilized adaptively to solve this nonlinear integer programming problem.
- **Simulation experiments** are designed and verify the rationality of the resource reallocation scheme, which provide the theoretical supports for the medical model .